Yuting Huang

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EDUCATION

University College London London, UK | Sep.2024-Sep.2025(Expected)
MSc Robotics and Artificial Intelligence, **the First Class Distinction(Expected)**
Relevant Courses: Machine Learning for Robotics, Estimation and Control, Computer Vision and Sensing, Modelling and Motion Planning, Object Detection and Classification, Introduction to Soft Robotics;

Heriot-Watt University Edinburgh, UK | Sep.2020-May.2024
Bachelor of Engineering in Robotics with Honours, **2:1 Degree(Grades: 69/100)**
Relevant Courses: Education of Creation and Innovation(100), Integrated Group Robotics Project(95), Robotics Systems Science(88), Introduction to Electrical and Electronic Engineering(86), Praxis Electronic Design(84), Introduction to Robotics(83);

Ocean University of China Qingdao, China | Sep. 2020-Jul. 2024
Bachelor of Computer Science and Technology, **Grades*: 82/100***
Relevant Courses: Advanced Mathematics, College Physics, Software Development(Java), Electronics and Circuits, Data Structure and Algorithm, Artificial Intelligence, Time frequency and signal analysis, Hardware software interface.

WORK EXPERIENCE

UK STEMAspire, *Mentee* London, UK | Jan.2025 - Present
- Participate in a female mentorship programme connecting students with mentors at Dell Technologies, where I have meetings with mentors at least once a month to shape my career path;
- Engage in career development sessions focused on empowering female students in the technology sector, including coaching on CVS, interviews, and networking skills.

Outlier AI, *Applied Mathematician - Chinese* Remote, UK | Dec.2024 – Feb.2025
- Assessed the factuality and relevance of domain-specific text produced by AI models;
- Crafted and answered questions related to Math;
- Evaluated and ranked domain-specific responses generated by AI models;

The National Robotarium, *HRI Student Research Assistant* Edinburgh, UK | Jan.2024-May.2024
I contributed to an EPSRC CDT research project in Human-Robot Interaction, focusing on designing an interactive Furhat robot dialogue system using GPT-3.5 and narrative-based direction-giving in Minecraft setup.
- Participated in weekly meetings to discuss system design, and was actively involved in drafting the ethics approval and designing 1 consent form, 1 pre-task form, 2 post-navigation forms and 2 follow-up question forms;
- Supported participant recruitment(24), questionnaire administration(144), time tracking, and note-taking during experiments, as well as data analysis to evaluate the impact of follow-up questions on instruction recall, usefulness, and accuracy;
- Acknowledged in the publication *"Follow the Yellow or Red Brick Road? Investigating the Impact of Narratives in a Guided Navigation Task"* (HAI 2024).

Heriot-Watt University, *EECE Offer Holder Day Student Ambassador* Edinburgh, UK | Apr.2024
- Acted as a point of contact and support for prospective students, offering personalised guidance and sharing insights into the student experience within the Robotics undergraduate Honours programme;
- Delivered a presentation on my final-year project to approximately 100 offer holders and their parents, addressing both technical and student life–related questions;
- Collaborated with fellow student ambassadors to lead campus tours, with a focus on Robotics laboratories and facilities, offering insights into ongoing experiments and research opportunities.

Shenzhen Hanyang Technology Co., LTD, *Embedded Software Development Engineer* Shenzhen, China | Jun.2022-Sep.2022
- Involved in Yarbo Snow Blower robots research and development, I conducted tests alongside senior robotics developers on various driveways, including large, steep, gravel, and circular roads;
- Developed and optimised software; edited around 200 lines of code in Python for smoother driving in different scenarios, and debugged to enhance the performance of the snow-blowing robot software;
- Authored robot navigation software modules and prepared technical documents detailing the software modules and their functionalities.

ACADEMIC PROJECTS (see link: Yukiyt778.github.io for more information)

Advanced Robotics and Remote Operations for Nuclear Environments London, UK | May.2025-Sep.2025
The final project for my master's degree focuses on developing advanced robotics, remote operations, and teleoperation techniques for challenging nuclear environments. Working in collaboration with the UK Atomic Energy Authority and AtkinsRealis, the aim is to improve the efficiency, safety, and accuracy of robotic operations used in decommissioning, maintenance, and fusion energy research. These operations often require robots to perform precise tasks using intelligent path planning for high degree of freedom(14 DoF)

robotics in hazardous or hard-to-access areas where human presence is limited or impossible. I'm responsible for optimising the path planning algorithms, considering age damping and other constraints, with practical work on KUKA robotic arm at HERE EAST.

Underwater Octopus-Inspired Gripper London, UK | Apr.2025 -May.2025

Designed and built an octopus-inspired underwater soft gripper using 3D printing and embedded electronics as part of my soft robotics module at the UCL Innovation Lab. The gripper has been tested in an underwater scenario with various shapes and weights of objects, including a ruler, keys, a tennis ball, and bottles. I received a mark of 100% for this project. I also gained experience in CAD design, casting, and laser cutting during the lab sessions for this course.

Pick-and-Place using myCobot Manipulator London, UK | Oct.2024-Nov.2024

Implemented pick-and-place functionality with a random colour cube randomly placed into a fixed bin on myCobot 280 Robotic Arm at the UCL Innovation Lab, utilising HSV filtering through a USB camera to differentiate between the object and background, and integrating ROS 2 for robotic arm manipulation.

Mechanical Design of a Robotic Waste Collection Mechanism Edinburgh, UK | Sep.2023 -May.2024

Designed and developed a robotic waste collection mechanism for train carriages using the Leo Rover platform. Utilised CAD design, 3D printing, and laser cutting to fabricate the structure, and integrated motors with brushes and electronics for actuation control from the GRID lab at Heriot-Watt University. Implemented autonomous navigation using ROS 1 and a trained waste detection model(SSD MobileNet v2) run on a Raspberry Pi 4B. Co-authored(2nd) the paper *"Development of a Waste Collection Robot for Train Carriages Interior"*, accepted at IEEE ARM 2025.

The Goblin's Challenge Edinburgh, UK | Sep.2023 – Nov.2023

Implemented an interactive game set in a D&D background on the Misty robot at The National Robotarium, focusing on gesture recognition using ResNet-50 and built-in storytelling functionality to enhance human-robot interaction.

Book Returning Robot in Library Qingdao, China | Feb.2023 -Jun.2023

Designed and developed a book-return robot for OUC Library using DJI Robomaster and MechArm, and independently built the bookshelves from scratch using wooden pieces. Programmed the software to implement line-following and obstacle-avoidance functionalities, ensuring efficient robot navigation.

Nanyang Technological University Summer Programme Online | Jul.2021-Aug.2021

Summer Programme: Robotics, Automation and Internet of Things

Awards: Distinction among individual assessment, group work, and presentation

I completed the online programme which comprised a series of lectures and discussions, as well as a group presentation. This programme helps university students from China foster professional skills and knowledge in Motion Systems and Planning in Robotics, Processes and Control in Industrial Automation, Synchronous and Asynchronous Communication and IOT's Network Interface, Analogue and Digital Sensors IOT's Network Interface, Stepwise and Continuous Motion Actuators IOT's Network Interface and Group Presentation. This programme was delivered in English online with 18 academic hours. The workload for 1 Academic Unit (AU) in NTU is equivalent to 13 hours.

Robotic Car Challenge Qingdao, China | Nov.2020 -Apr.2021

Soldered an Arduino board along with an LED circuit designed by HWU and additional electronics on veroboards. The board was then integrated into a robotic car to implement line following(both straight lines and curves), light following and obstacle avoidance using an L298 motor driver board, LDR, IR, ultrasonic sensors and other electronics.

PUBLICATIONS & AWARDS

2025 10th IEEE International Conference on Advanced Robotics and Mechatronics, *Second Author* May.2025

Development of a Waste Collection Robot for Train Carriages Interior

2023 International Conference on Computing and Pattern Recognition, *Third Author* Oct.2023

A Method for Breast Mass Segmentation using Image Augmentation with SAM and Receptive Field Expansion

2023 MCM/ICM Mathematical Contest In Modeling, *Team Member* Jul.2023

Honorable Mention (Top 15%);

TECHNICAL SKILLS

Laboratory Techniques: CAD design, 3D printing, laser cutting, PCB fabrication and soldering.

Programming Languages: Python (Advanced), Java (Advanced), C/C++ (Intermediate), Shell (Intermediate).

Software & Tools: VS Code, Eclipse, Anaconda, git, PyTorch, MATLAB, LaTex.

Languages: Chinese Mandarin (Native), English (Advanced), Chinese Cantonese (Intermediate).